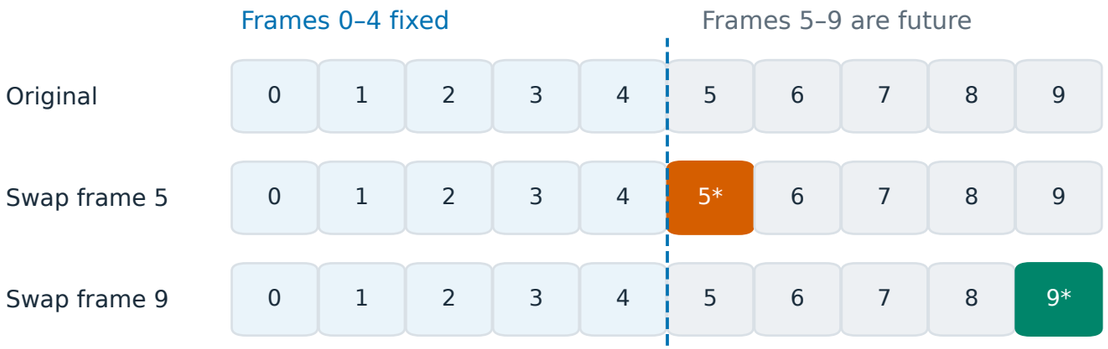
# Future frames change an earlier prediction

Frozen reference predictive coder | original Minecraft validation PNGs | Oscar 8098889

## a The intervention: change only the future

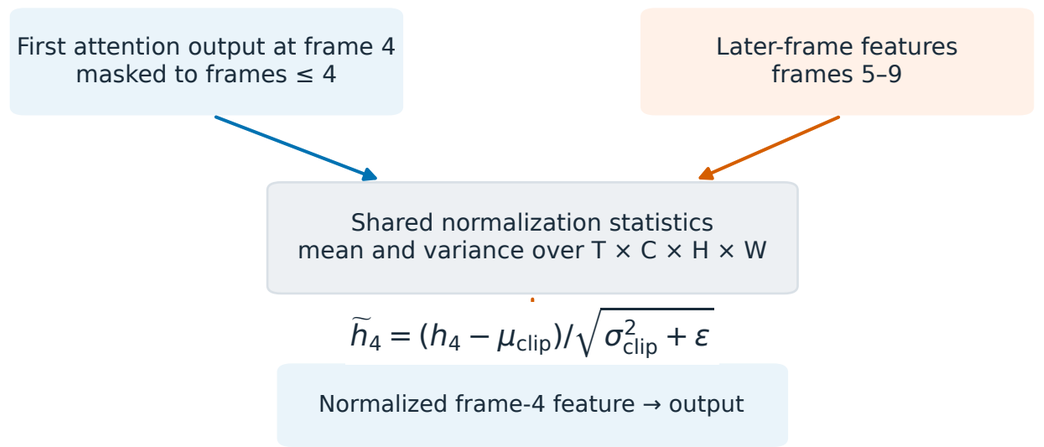



* Same-index frame from a second original episode.

Read out the same index each time: prediction $\hat{y}_5$ and latent $z_4$

Frozen weights • eval mode • identical repeat as a numerical control

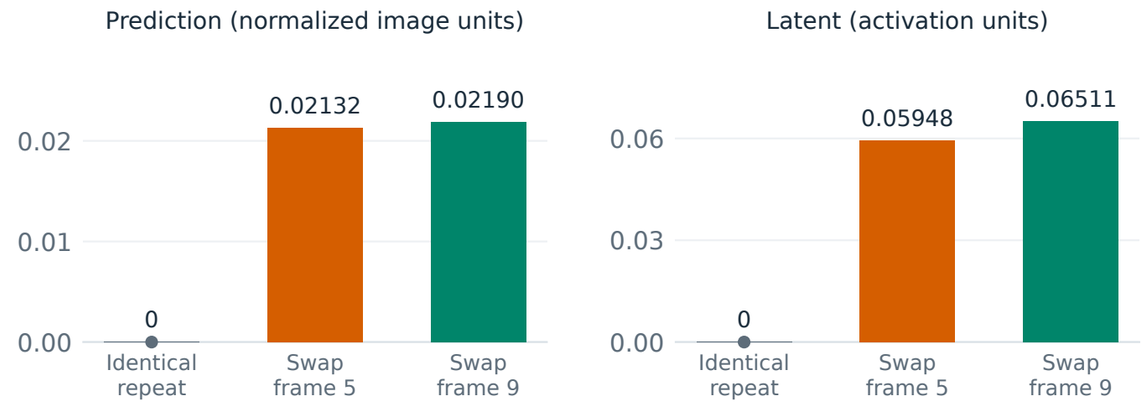
## b How future information can bypass the mask




Source-supported mechanism; normalization has not been ablated.

## c Measured dependence on future frames

Mean absolute change relative to the original input




A causal output must remain unchanged when only its future changes.
One clip + one donor; descriptive values, no error bars.
Image-output and latent changes have different units.

## d What follows — and what does not

**Confirmed**

The tested frame-4 output depends on future input.
The earlier 0.01293 RGB MSE is not a causal forecasting score.

**Not established**

How much leakage improved that MSE; whether spatial maps exist; or whether every evaluation in the paper is affected.

**Prepared spatial extraction**

Keep only $z_9$ from a window ending at frame 9.
No future input at that endpoint; this is not retraining or a full replication.

Evidence: Oscar 8098889, reported 29 Oct 2026. Reference commit fed80b72925a; frozen PredictiveCoder, two heads per attention module.

Frame tiles are schematic. Changes are measured on the same output index after a single future-frame substitution. Model source: jgornet/predictive-coding-recovers-maps.